This submission is with an approved extension

Assignment 3

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For each of the sections below, your reported test accuracy should approximately match the accuracy reported on Kaggle. It is fine to have a small deviation. Report both accuracies if they are different.

Self-supervised Learning on CIFAR10

1) Rotation training

Report the hyperparameters you used to train your model. Discuss any particular implementation choices which caused significant performance increases.
lr=0.001, 45 epochs, batch size 128
Adam optimizer with lr=0.001 caused significant performance increases

2) Fine-tuning late layers
Report the hyperparameters you used to fine-tune your model. Compare the performance between pre-trained model and randomly initialized model.
lr=0.0005
Pre-Trained Model: acc 67%
Randomly Initialized Model: acc 47%
Pre-Trained Model is better.

3) Fully supervised learning
Report the hyperparameters you used to fine-tune your model. Compare the performance between pre-trained model and randomly initialized model. Discuss anything you find interesting comparing fine-tuning the late layers only in section (2) and fine-tuning the whole model in section (3).
lr=0.0005, 20 epochs
Pre-Trained Model: acc 83%
Randomly Initialized Model: acc 82%
The two perform similarly.

4) With a more advanced architecture
Report the details of your architecture and the hyperparameters used to train this model. Compare the performance between this model and the models in sections (1) and (3).

5) Extra credit